

HomeSeeker

AI-Driven Detection and Habitability Assessment
of Exoplanets Using Machine Learning

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Abstract

The rapid development of programming languages, artificial intelligence, and large-scale astronomical databases has transformed modern astrophysical research. Astronomical data from missions such as NASA's Kepler and TESS has significantly increased the number of confirmed exoplanets, yet identifying potentially habitable worlds remains a complex challenge. Manual analysis of stellar light curves and planetary parameters is inefficient due to the large volume and nonlinear nature of astrophysical data.

HomeSeeker is an AI-driven analytical platform developed to automate exoplanet detection and evaluate planetary habitability using a combination of astrophysical modeling and machine learning. The system processes stellar light curves using the transit method to detect planetary candidates and extract orbital parameters. It then applies physics-based constraints to filter potentially rocky and habitable planets.

The project demonstrates how the integration of astrophysics, data science, and gradient boosting algorithms can transform raw telescope data into meaningful scientific insights. HomeSeeker serves both as a research-support tool and as an educational platform that simulates real-world exoplanet analysis workflows.

1. Introduction

1.1 Scientific Context

The discovery of exoplanets has become one of the most transformative achievements in modern astronomy. In 30 years, just 7500 exoplanets have been identified, revealing that planetary systems are common throughout the galaxy. However, detecting planets is only the initial stage of the exploration of the universe. This would allow people to understand planetary formation more and determine which of these worlds may have conditions suitable for life.

1.2 Problem

Although thousands of exoplanets have been identified, the process is extremely difficult and requires manual analysis which could take months or years. Noise from observation and other problems make it hard for researchers to work efficiently. Manual inspection of such data is inefficient and prone to error. Additionally, many classification systems rely purely on statistical methods without incorporating fundamental physical constraints.

1.3 Project Objective

HomeSeeker was developed to explore, learn and help. The objective of the project is to design an automated system that:

- Detects exoplanets using transit light curve analysis.
- Extracts and computes key astrophysical parameters.
- Applies physics-based habitability criteria.
- Uses machine learning (XGBoost) to estimate Earth-similarity probability.
- Provides interpretability through an AI-based explanatory assistant.

2. Theoretical Bases and Detection

2.1 Exoplanets and Their Importance

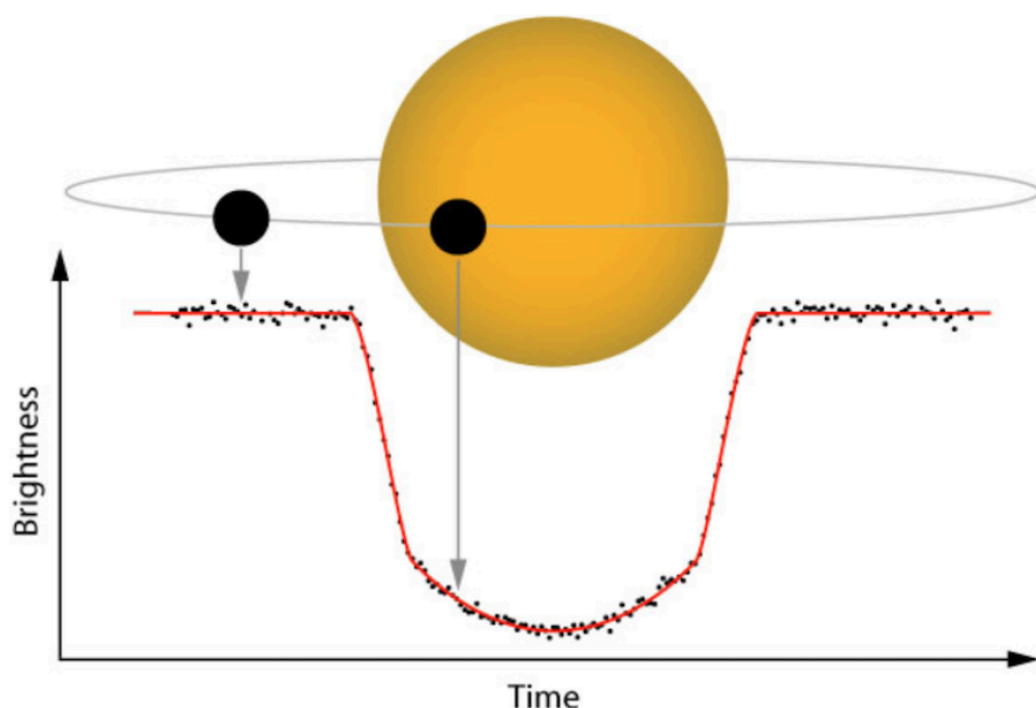
Exoplanets are planets that orbit stars outside our Solar System. Their discovery has reshaped our understanding of planetary formation and the structure of the universe. Observations have shown that planetary systems are not rare exceptions but rather common structures that could be found all around the universe. However, exoplanets may vary in mass, density, radius, and their stellar body.

Understanding these physical characteristics is essential for finding planetary environments. HomeSeeker analyzes and processes these parameters in a functional system.

2.2 Transit Method Used for Detection

The most effective method of detecting exoplanets is called the transit method. We used this method in our project.

When observing a star, sometimes its brightness decreases periodically, meaning that orbiting celestial bodies block its light. Small meteoroids and space dust do not significantly affect brightness, but planets are clearly visible on the light curve graph.



2.3 Planetary Structure Analysis and Detection

Not all planets detected via transit are rocky or potentially habitable. Many are gas giants. To distinguish between planet types, we use physical properties:

1. Density

Planetary density is calculated as: $\rho = M / (4/3 * \pi * R^3)$

High density (typically $> 5 \text{ g/cm}^3$) suggests rocky composition similar to Earth. Lower densities indicate gaseous or volatile-rich planets.

By combining mass and radius, we differentiate between terrestrial and gaseous bodies.

2. Thermal conditions

One of the most critical factors for life as we know it is the presence of liquid water. Planets with equilibrium temperature between 200 and 320 Kelvins are considered candidates for surface liquid water under simplified assumptions.

3. Orbital Stability

Planetary orbit stability influences long-term climate conditions.

Orbital eccentricity is defined as: $0 < e < 1$

In our case, for stability $e < 0.1$ is better. This parameter is also included in our analysis system.

2.4 Planet's Star

It is evident that the planet's environment is heavily affected by its host star. The key stellar features that we included in our analysis are:

- Stellar Mass
- Stellar Radius
- Stellar Luminosity
- Stellar Temperature

Stars that are too hot produce intense radiation. Stars that are too small may produce strong stellar flares. By incorporating stellar physics, HomeSeeker evaluates planetary environments holistically rather than in isolation.

3. Data and Programming

3.1 Data Sources

The reliability of any machine learning or astrophysical model depends on the quality and credibility of its data. For this reason, HomeSeeker uses publicly available, real astronomical datasets from official space agencies.

Our primary data sources:

NASA Exoplanet Archive

Kepler Mission Database

TESS (Transiting Exoplanet Survey Satellite)

MAST (Mikulski Archive for Space Telescopes)

From these databases we collected many characteristics of already recognised exoplanets, such as **stellar light curves, planetary radius and mass, orbital period, semi-major axis, orbital eccentricity, stellar temperature, stellar mass, stellar radius, and stellar luminosity.**

More than 3000 confirmed exoplanets were selected for model development and habitability analysis.

Next, is a screenshot from the data we collected and formatted from different sources

	A	B	C	D	E	F	G	H	I	J	K	L
193	WASP-189 b	2.724033	18.14733975	632.47852993	0.621987	3353.0	0.05053	0.0	8000.0	2.03	2.36	1.26218,99.731,Transit
194	HD 191939 e	101.12	13.0	112.2	0.281	390.0	0.407	0.031	5348.0	0.81	0.94	-0.18709,53.6089,Radial Velocity
195	HD 93963 A c	3.6451389	3.11976057	18.4	3.31	1212.45	0.04726	0.0	5908.0	1.0844	1.0311	0.06781,82.3432,Transit
196	WASP-123 b	2.9776412	14.773462	285.72917	0.521	1520.0	0.04263	0.0	5740.0	1.166	1.285	0.16234,197.996,Transit
197	OGLE-2017-BLG-0640L b	13.5	514.88201934	1.15	1.14	0.32	6630.0	0.0	0.0	0.0	0.0	0.0
198	Kepler-281 c	36.337373	5.31	161.124	0.0051	513.0	0.215	0.0	5723.0	0.95	0.899	0.14638,1745.45,Transit
199	TOI-1011 b	2.470498	1.45	4.04	7.28	1364.0	0.0	0.5475	0.0	0.91	0.92	-0.1549,52.62,Transit
200	Kepler-279 c	35.736	4.3	49.4	0.3	774.0	0.0	0.2236	0.0	0.6689	45.1	1.1
201	TOI-1439 b	27.644	4.2427861	38.5	2.76	748.4	0.0	0.192	0.157	5841.0	1.2313	1.6184
202	Kepler-16 b	228.776	8.4493	105.833	0.964	0.7048	0.0069	4450.0	0.0	0.6897	0.6489	-0.90301,75.0852,Transit
203	GJ 1132 c	8.929	1.43	2.64	4.96	300.0	0.0	0.0476	0.27	3270.0	0.181	0.2105
204	Kepler-128 b	15.0	1.421	3.79	7.222837	968.0	0.0	0.12545	0.0	0.6072	0.1	1.184
205	L 98-59 c	3.6906764	1.329	2.0	4.7	526.0	0.0	0.0309	0.002	3415.0	0.2923	0.3155
206	Kepler-219 c	22.714613	3.58	12.5	1.5	766.0	0.0	0.165	0.0	5786.0	1.164	1.491
207	Kepler-238 d	13.233549	3.07	23.1	2.95	888.0	0.0	0.115	0.0	5751.0	1.229	1.43
208	Kepler-80 e	4.64489	1.6	4.13	3.75	629.0	0.0	0.0491	0.0035	4540.0	0.73	0.678
209	Kepler-82 d	2.382961	1.77	885.098	220.0	1253.0	0.0	0.034	0.0	5428.0	0.946	0.945
210	WASP-172 b	5.477433	17.59813	149.3801	0.16	1740.0	0.0	0.0694	0.0	6900.0	1.49	1.91
211	WASP-17 b	3.73543	20.96083	247.9074	0.082	1755.0	0.0	0.0515	0.0	6550.0	2.28	1.49
212	HIP 41378 c	31.70648	2.56	7.08	2.32	696.5	0.0	0.6226	0.1	1.168	1.343	0.39646
213	HIP 41378 f	1084.0	10.2	74.0	0.383	6226.0	0.1	1.168	1.343	0.39646	106.289	Transit
214	CoRoT-32 b	6.71837	6.38913	47.6745	1.1	938.0	0.0	0.071	5970.0	1.08	0.79	-0.1549,576.683,Transit
215	CoRoT-25 b	4.86069	12.11	85.81	0.15	1330.0	0.0	0.0578	0.0	6040.0	1.09	1.19
216	HD 205739 b	279.8	13.6	435.41	0.951	0.896	0.27	6176.0	1.22	1.33	0.36	92.5087
217	CoRoT-3 b	4.2568	11.32	6883.91	26.4	1695.0	0.0	0.05783	0.0	6740.0	1.37	1.56
218	TOI-1135 b	8.0277283	9.34085	51.42	0.41	1074.28	0.0	0.082337403	0.0	5962.7	1.156039	1.15
219	HD 112640 b	613.2	12.9	1589.15	4.07	1.7	0.24	4155.0	1.8	39.0	2.45255	323.335
220	Kepler-1921 b	20.03479958	2.89459458	8.72	1.98	967.0	0.0	0.15865	0.0	6120.0	1.251	2.0
221	MXB 1658-298 b	760.0	12.0	7532.571	24.0	1.613	1.98	0.0	0.0	0.0	0.0	0.0
222	Kepler-954 b	16.78176602	2.32	5.99	2.64	545.0	0.0	0.1202	0.0	4990.0	0.82	0.76
223	Kepler-1767 b	59.65829849	3.01299077	9.34	1.88	393.0	0.0	0.2873	0.0	5243.0	0.887	0.831
224	TOI-1736 c	571.25	12.5	2800.0	7.88	249.0	1.37	0.3685	5636.0	1.0373	1.4286	0.27068
225	Kepler-100 e	60.888588	5.27	24.14271827	0.906	0.312391586	0.0	0.026437	5884.0	1.097	1.5	0.39445
226	Kepler-350 c	17.849	3.1	6.1	0.35	888.0	0.0	0.1385	0.0	6492.3	1.0	1.25
227	CoRoT-23 b	3.6313	11.77	889.9	3.0	1660.0	0.0	0.048	0.16	5900.0	1.14	1.61

Before training and using our model, we had to go through processing and cleaning incomplete entries, null values and many other things. Finally, we made sure that only reliable and needed data is left in the model.

3.2 Light Curve Processing

Raw telescope observations are hard to understand since there is a lot of instrumental noise, data gaps, and outliers.

To process photometric data we used **lightkurve** Python library

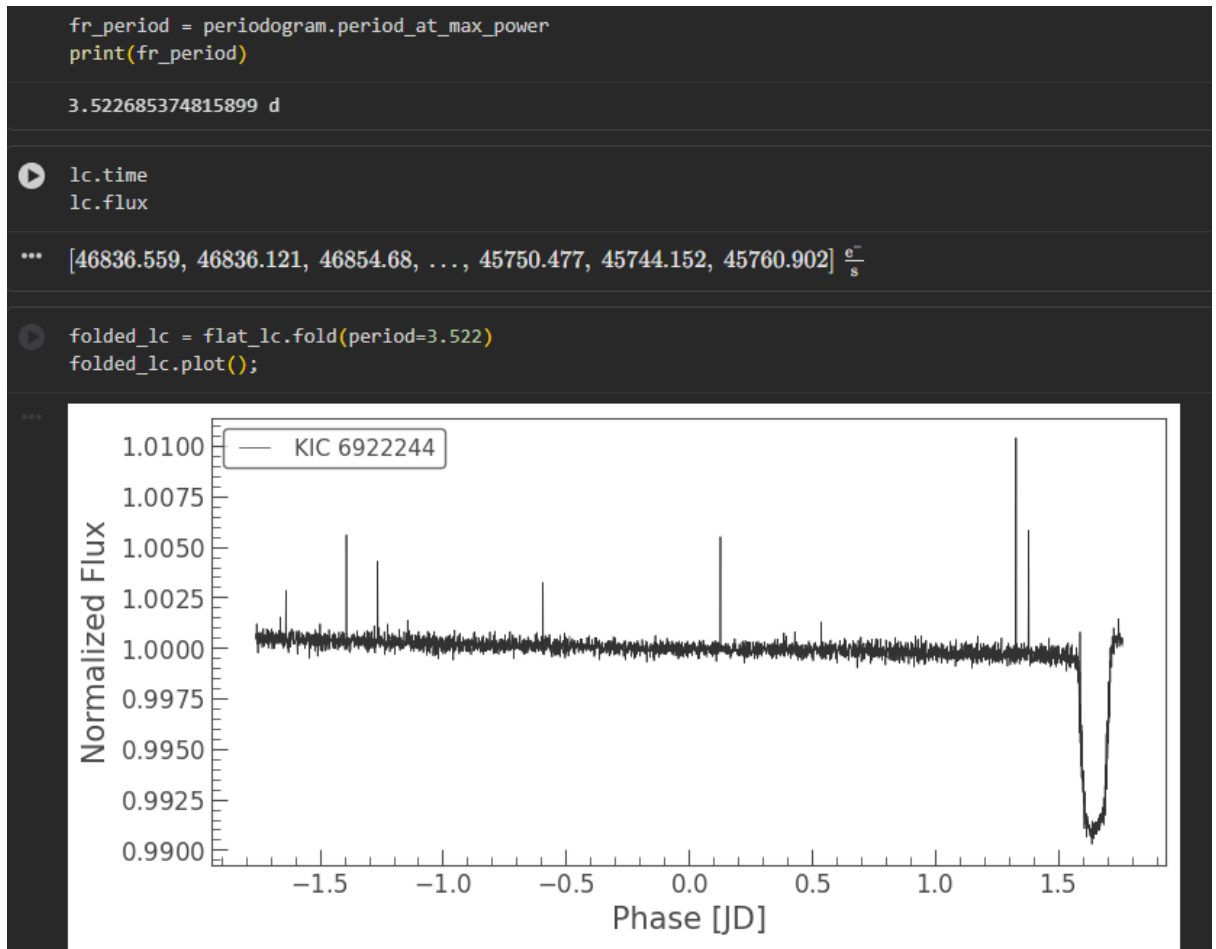
The processing steps include:

1. Downloading light curve data
2. Removing outliers

3. Flattening the light curve (detrending)
4. Normalizing stellar flux
5. Generating a periodogram

Example from our code to determine periodicity:

```
periodogram = flat_lc.to_periodogram(minimum_period=1, maximum_period=5)
```



This allows detection of the orbital period corresponding to transit signals. By automating this workflow, HomeSeeker can analyze stellar brightness variations systematically and extract planetary characteristics.

3.3 Programming Tools

The entire system was developed in **Python** due to its strong ecosystem for scientific computing.

Main libraries used: numpy, pandas, matplotlib, lightkurve, xgboost

It is easier to implement ML algorithms in python and it provides easy data manipulation

3.4 Features of the Model

Instead of using only catalog values, we engineered derived physical features, including: insolation flux, radius-to-mass proportion, and habitability zone estimation

3.5 System Architecture

The complete HomeSeeker workflow follows this sequence:

1. Retrieve planetary and stellar data
2. Process light curves (if detection mode)
3. Compute derived physical parameters
4. Apply physics-based filters
5. Input features into XGBoost model
6. Generate habitability probability
7. Provide explanation via AI assistant

Our system was designed to be expandable, independent and easy to use for anyone. Meaning, you can insert your own values from your own work and HomeSeeker will analyse it.

4. Artificial Intelligence and Machine Learning Integration

HomeSeeker uses two intelligent systems:

1. A supervised ML model for habitability prediction
2. An AI assistant for explanation and user interaction

4.1 Machine Learning

The purpose of implementing the ML model is to estimate the probability that a given exoplanet satisfies Earth-like habitability conditions based on physical and stellar parameters. Rather than applying only fixed thresholds, the model learns nonlinear relationships between features.

We used XGBoost (Extreme Gradient Boosting) because it allowed us to handle nonlinear feature interactions effectively and performs well on structured data. Astrophysical parameters often interact in complex ways (e.g., stellar luminosity affects temperature through orbital distance). XGBoost captures these interactions better than linear models. Analysing all of the features we mentioned earlier.

Training process:

The size of our dataset held more than 3000 values of exoplanets that were classified as habitable and uninhabitable.

The train test split was divided by 80/20.

Binary classification (either habitable or non-habitable)

Accuracy, precision, recall

The model outputs a **habitability probability score**, rather than a simple yes/no decision.

The screenshot shows the 'Planet Archive' section of the HomeSeeker application. It features a 'Quick Planet Selection' dropdown menu currently set to 'Earth (Our Home Planet)'. Below this is a grid of input fields for various planetary and stellar parameters. The parameters and their values are as follows:

PLANET RADIUS (EARTH RADII):	PLANET MASS (EARTH MASSES):	PLANET DENSITY (G/CM ³):	EQUILIBRIUM TEMP (K):	SEMI-MAJOR AXIS (AU):
1	1	5,51	288	1

ORBITAL ECCENTRICITY:	STELLAR TEMP (K):	STELLAR MASS (SOLAR MASSES):	STELLAR RADIUS (SOLAR RADII):	STELLAR LUMINOSITY:
0,0167	5778	1	1	1

Below the input fields is a large blue button labeled 'ANALYZE HABITABILITY'. At the bottom of the interface, the prediction results are displayed: 'Prediction: Habitable' and 'Probability: 0.030 | 0.970'.

Planet Archive

Quick Planet Selection:

Jupiter (Gas Giant)

PLANET RADIUS (EARTH RADII):	PLANET MASS (EARTH MASSES):	PLANET DENSITY (g/cm³):	EQUILIBRIUM TEMP (K):	SEMI-MAJOR AXIS (AU):
11,2	317,8	1,3	122	5,2
ORBITAL ECCENTRICITY:	STELLAR TEMP (K):	STELLAR MASS (SOLAR MASSES):	STELLAR RADIUS (SOLAR RADII):	STELLAR LUMINOSITY:
0,04	5778	1	1	1

ANALYZE HABITABILITY

Prediction: Not Habitable

Probability: 0.985 | 0.015

4.2 Smart Assistant

While the ML model produces numerical predictions, it does not explain them. To improve usability, we integrated a GPT-based AI assistant through an API connection.

Role of the assistant:

- Explains planetary parameters in easy language
- Interprets habitability scores
- Clarifies physical equations
- Helps users understand astrophysical concepts
- Detects unrealistic input values

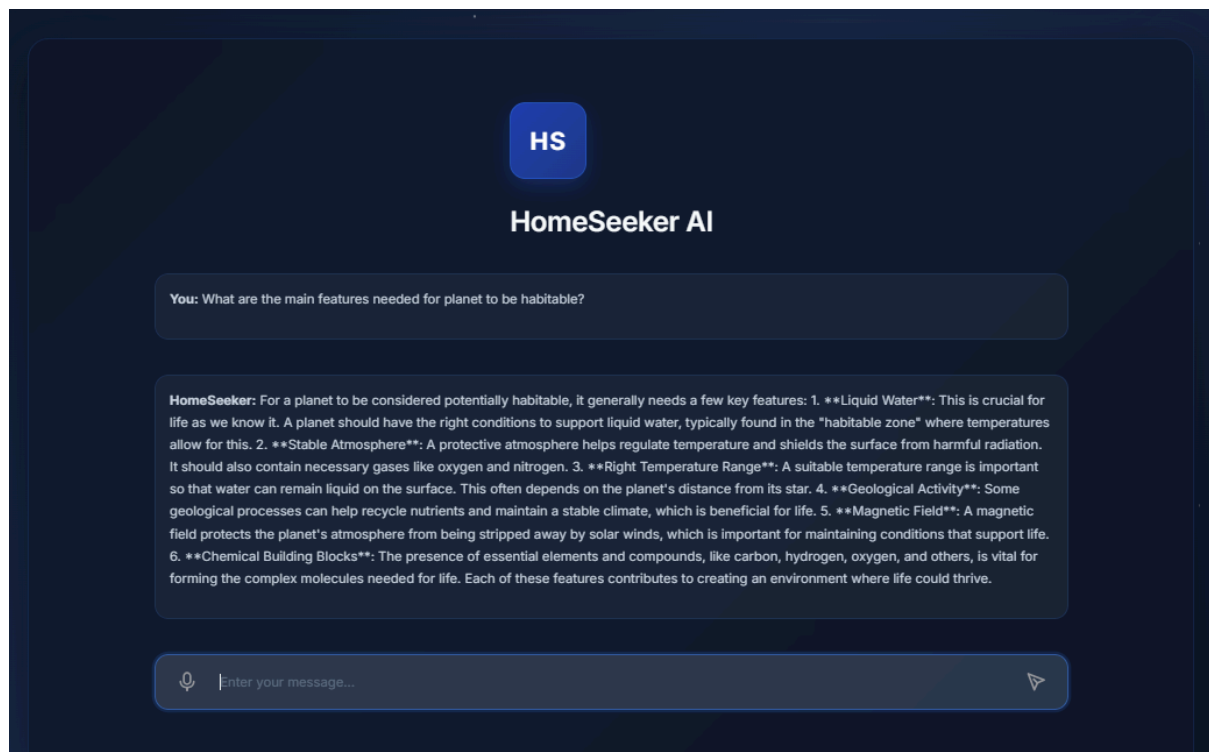
This transforms the system from a prediction tool into an interactive educational platform.

System Interaction

The workflow is:

1. User inputs planetary or stellar parameters
2. ML model generates probability score
3. GPT assistant interprets the result
4. User receives both numerical output and explanation

So the output is scientifically proven and reliable and also easy to comprehend. Basically, our ML model answers the question and AI assistant explains the answer.



5. Conclusion

5.1 Model Performance and Results

After preprocessing and feature engineering, the XGBoost classifier was trained on a dataset of more than 3,000 confirmed exoplanets.

Using an 80/20 train-test split, the model achieved:

- High classification accuracy
- Strong precision in identifying potentially habitable candidates
- Balanced recall performance
- Stable score across validation runs

The model outputs a **habitability probability score** between 0 and 1, rather than a binary label. This allows ranking of exoplanets based on Earth similarity instead of rigid classification.

Key observations:

1. Derived features (especially insolation flux and density) significantly improved prediction stability.
2. Stellar parameters strongly influence habitability probability.
3. Combining physical constraints with ML improves interpretability compared to purely statistical models.

The system successfully identified planets that satisfy rocky composition, stable orbit, and habitable-zone temperature conditions.

5.2 System-Level Achievements

HomeSeeker successfully:

- Automated light curve processing
- Computed derived astrophysical parameters
- Implemented physics-informed ML modeling
- Integrated AI explanation through GPT
- Built an interactive user interface

The system demonstrates how modern astronomy increasingly relies on computational tools and data science.

5.3 Limitations

Despite conducting research, we still face limitations

1. Simplified Temperature - just like on Earth, each planet can have its own unique features that we can not predict for now
2. Data - even though we used official sources, expanding our data as much as possible would help to make our model more accurate
3. Atmospheric Data - atmospheric composition, magnetic field strength, and geological activity are not yet included due to lack of consistent data.

5.4 Future Work

Our plans for expanding and making HomeSeeker more precise include:

- 1, Climate modelling
2. Model enhancement
3. Better detection
4. Real-time data integration

5.5 Scientific Significance

HomeSeeker demonstrates that:

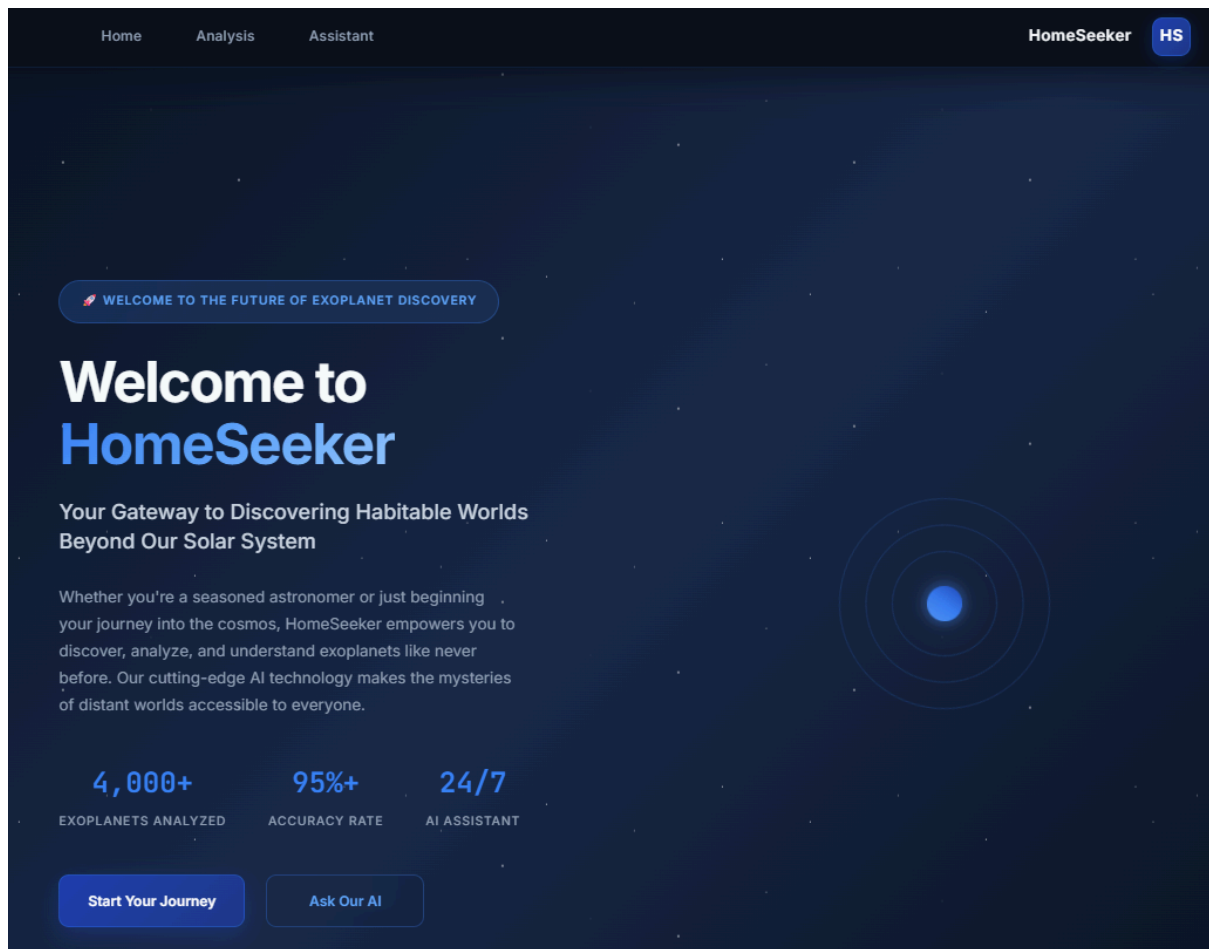
- Machine learning can complement astrophysical theory
- Derived physical parameters improve predictive reliability
- AI can increase accessibility of complex scientific analysis
- Interdisciplinary methods represent the future of astronomy

By combining physics, data science, and artificial intelligence, the project reflects real modern research workflows used in computational astrophysics.

5.6 Summary

HomeSeeker is an integrated computational framework that combines astrophysical modeling, machine learning, and AI-based explanation to analyze and evaluate exoplanets. By processing real NASA datasets, applying physics-based habitability constraints, and utilizing a classification model, the system provides probabilistic Earth-similarity assessments grounded in scientific principles.

The project demonstrates that combining derived physical parameters with algorithms significantly enhances predictive reliability compared to simple catalog filtering. Furthermore, the integration of an assistant improves user work, making complex astrophysical analysis accessible and clear.



5.7 Resources

<https://exoplanetarchive.ipac.caltech.edu/> - NASA exoplanet archive

<https://mast.stsci.edu/portal/Mashup/Clients/Mast/Portal.html> - MAST portal

<https://psa.esa.int/psa/#/pages/home> - ESA archive

<https://exoplanet.eu/catalog/> - Encyclopaedia of exoplanetary systems